Kondo and RKKY interactions in bilayer graphene

Aravindh Swaminathan Shankar

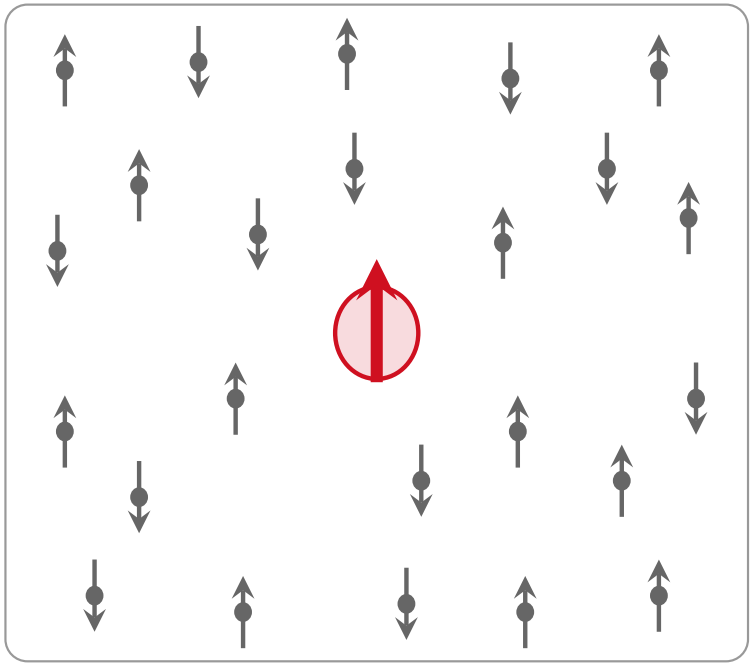
ashankar@ictp.it

with D. O. Oriekhov, L. Fritz, A. K. Mitchell

Abdus Salam ICTP, Trieste, Italy

NM4QM 2026

A magnetic impurity in a metal is screened away below T_K: the Kondo effect

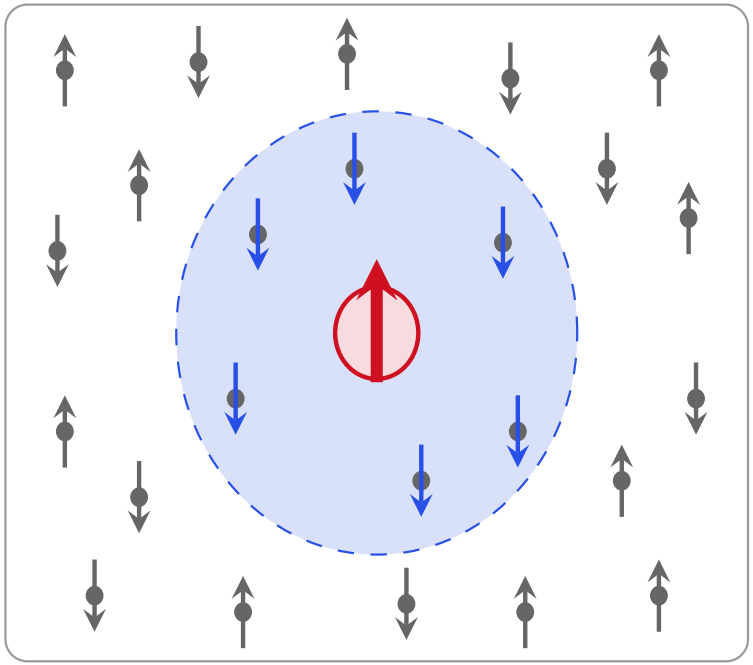


local moment

$$H = H_{\text{host}} + J\,\vec{S}\cdot\vec{s}(0)$$

Impurity spin coupled to the conduction electrons

A magnetic impurity in a metal is screened away below T_K: the Kondo effect



Kondo singlet

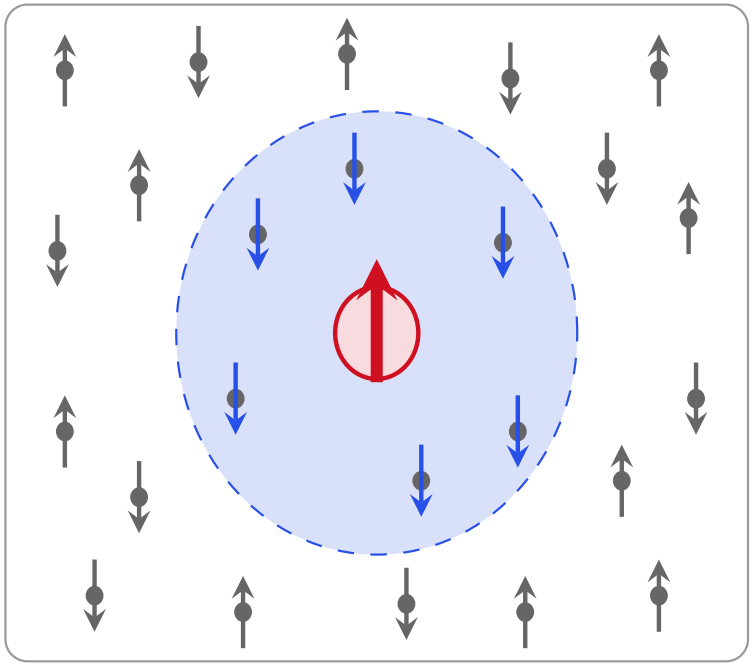
$$H = H_{\text{host}} + J\,\vec{S}\cdot\vec{s}(0)$$

Impurity spin coupled to the conduction electrons

Low T: the electrons rearrange and screen the moment

Strong coupling: beyond perturbation theory

A magnetic impurity in a metal is screened away below T_K: the Kondo effect



Kondo singlet

$$H = H_{\text{host}} + J\,\vec{S}\cdot\vec{s}(0)$$

Impurity spin coupled to the conduction electrons

Low T: the electrons rearrange and screen the moment

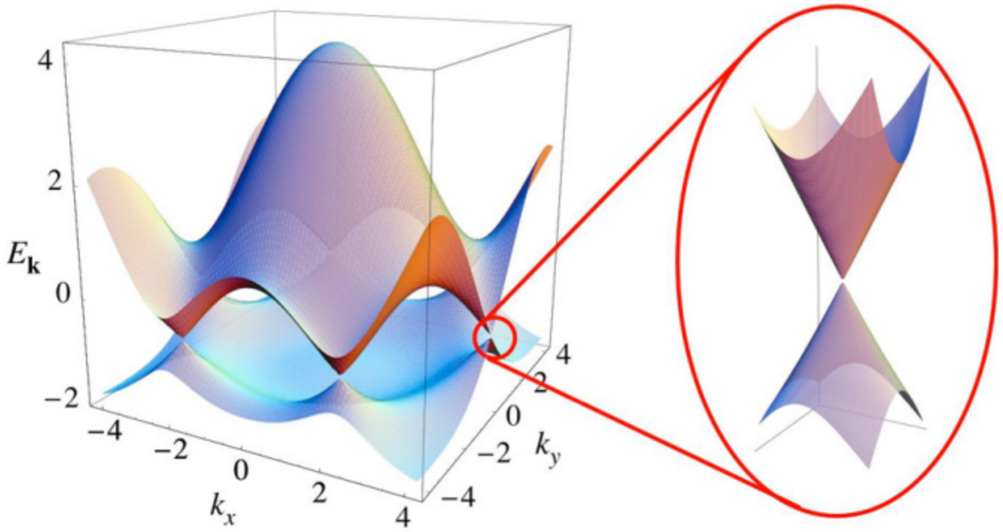
Strong coupling: beyond perturbation theory

$$T_K = D\,e^{-1/\rho_0 J}$$

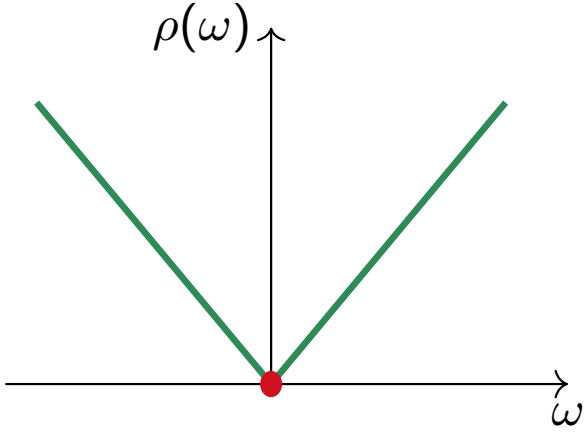
host density of states at the Fermi level

The host density of states decides if, and when, the moment is screened

No Kondo effect in graphene at charge neutrality



Castro Neto et al., Rev. Mod. Phys. 81, 109 (2009)



$\rho(E_F) = 0$

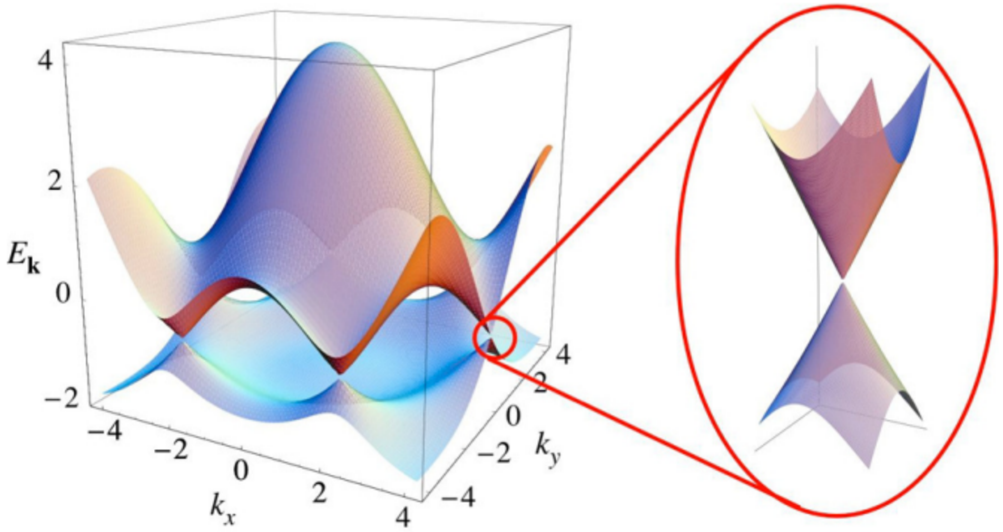
$$T_K = D\, e^{-1/\rho_0 J}$$

$\rho_0 \to 0 \;\Rightarrow\; T_K \to 0$

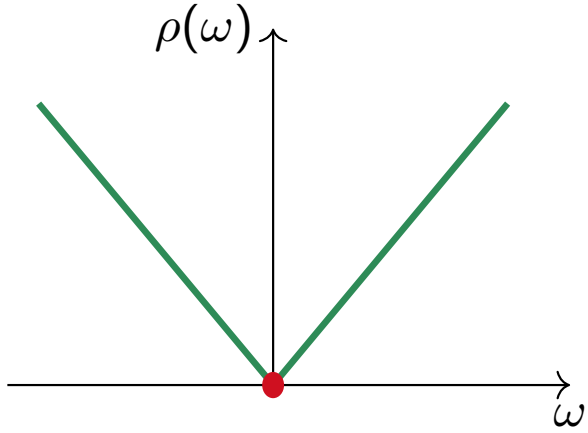
the moment stays free

What happens in other graphene-like materials?

No Kondo effect in graphene at charge neutrality



Castro Neto et al., Rev. Mod. Phys. 81, 109 (2009)



$\rho(E_F) = 0$

$$T_K = D\, e^{-1/\rho_0 J}$$

$\rho_0 \to 0 \;\Rightarrow\; T_K \to 0$

the moment stays free

Experiments in graphene

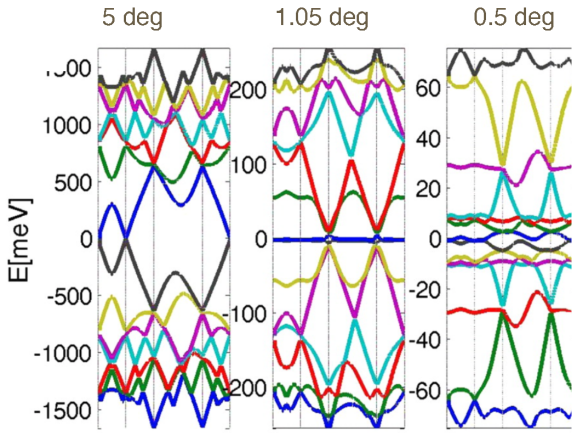
H adatoms: González-Herrero et al., Science 352, 437 (2016)
Vacancies: Jiang, . . . , Andrei, Nat. Commun. 9, 2349 (2018)

What happens in other graphene-like materials?

Model: Anderson impurity on twisted bilayer graphene

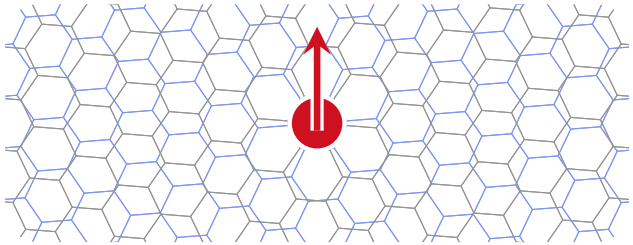
Host Bistritzer–MacDonald continuum model

$$H(\mathbf{k}) = \begin{pmatrix} h^K_{\theta/2}(\mathbf{k}) & wT_1 & wT_2 & wT_3 \\ wT_1^\dagger & h^K_{-\theta/2}(\mathbf{k}-\mathbf{q}_1) & 0 & 0 \\ wT_2^\dagger & 0 & h^K_{-\theta/2}(\mathbf{k}-\mathbf{q}_2) & 0 \\ wT_3^\dagger & 0 & 0 & h^K_{-\theta/2}(\mathbf{k}-\mathbf{q}_3) \end{pmatrix}$$



Bistritzer and MacDonald, PNAS 108, 12233 (2011)

Impurity Anderson model



$$H_{\text{imp}} = \epsilon_d \sum_\sigma n_{d\sigma} + U\, n_{d\uparrow} n_{d\downarrow} + V \sum_\sigma \left(d_\sigma^\dagger c_{0\sigma} + \text{h.c.} \right)$$

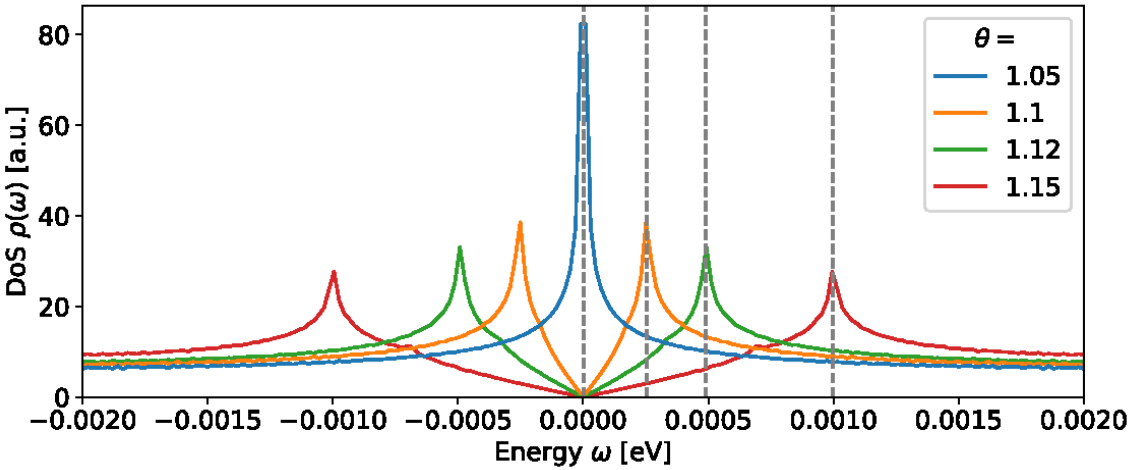
$|0\rangle$ $|\uparrow\rangle$ $|\downarrow\rangle$ $|\uparrow\downarrow\rangle$

4 states; large U leaves $\uparrow, \downarrow$:
a spin-$\frac{1}{2}$ moment, $J \sim V^2/U$

Impurities in TBG: Shankar, Oriekhov, Mitchell, Fritz, PRB 107, 245102 (2023);

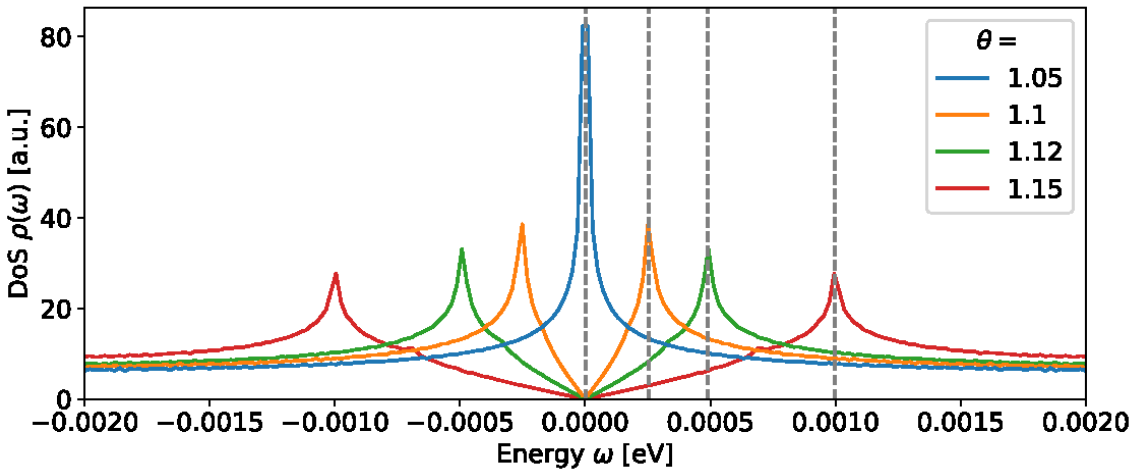
Chang, Yi, Wu, Kugler, Andrei, Vanderbilt, Kotliar, Pixley, PRL 133, 126503 (2024)

van Hove singularities in the flat band



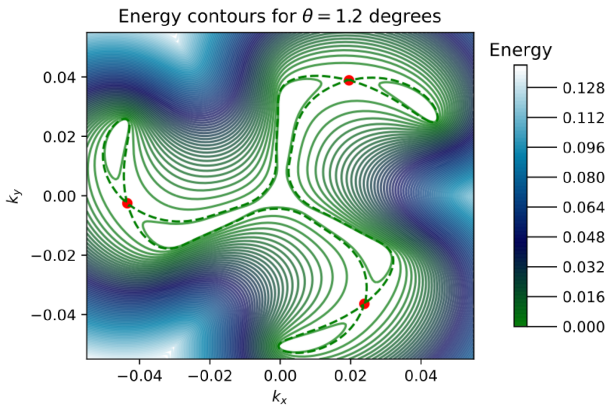
Dirac cone at $\omega = 0$: $\rho \sim |\omega|$

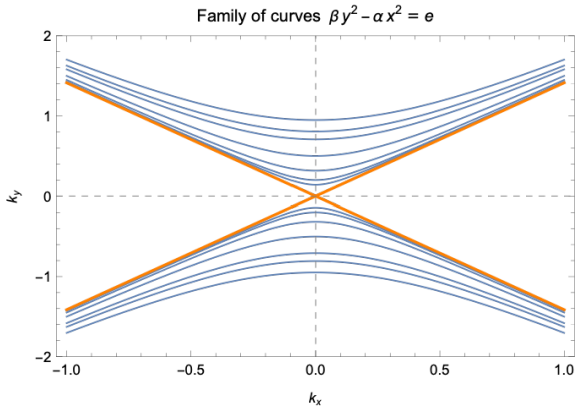
van Hove singularities in the flat band



Dirac cone at $\omega = 0$: $\rho \sim |\omega|$

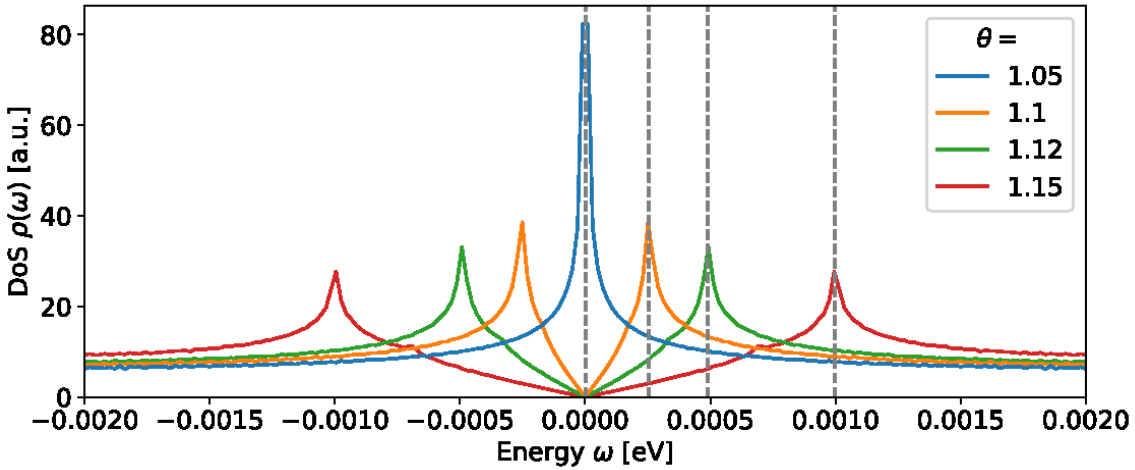
van Hove peaks at $\pm E_v$





ordinary saddle $\quad E = E_v + \alpha p_x^2 - \beta p_y^2 \implies \rho(E) \sim \log \left| \frac{\Lambda}{E - E_v} \right|$

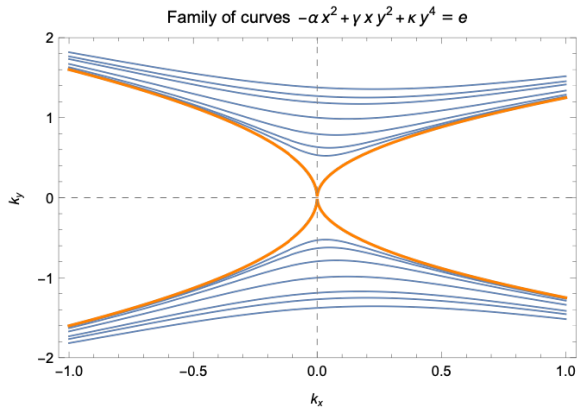
van Hove singularities in the flat band

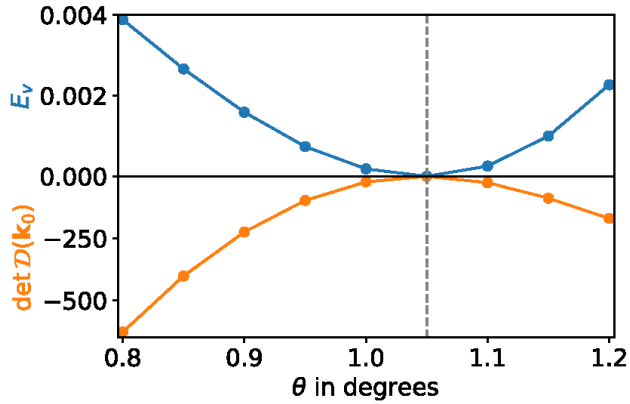


Dirac cone at $\omega = 0$: $\rho \sim |\omega|$

van Hove peaks at $\pm E_v$

1.05°: the peaks merge





higher-order saddle $\quad E = \alpha p_x^2 + \gamma p_x p_y^2 + \kappa p_y^4 \implies \rho(E) \sim |E|^{-1/4}$

Yuan, Isobe, Fu, Nat. Commun. 10, 5769 (2019)

At 1.05° the two vHs merge into a higher-order vHs

A van Hove peak boosts T_K

	Density of states $\rho(\omega)$	T_K	$S_{\rm imp}(T \to 0)$
Metal	ρ_0 : constant	$D\, e^{-1/\rho_0 J}$	0
Dirac cone	$\rho_0 \lvert\omega\rvert$: linear pseudogap	$T_K = 0$	$\ln 2$
Standard van Hove	$\rho_0 \left[1 + a \ln(D/\lvert\omega\rvert)\right]$: log divergence	$D\, e^{\frac{1}{a} - \frac{1}{a}\sqrt{1+\frac{2a}{\rho_0 J}}}$	0
Higher-order van Hove	$\rho_0 \lvert\omega\rvert^{-\alpha}$: power-law divergence	$D \left(1 + \frac{\alpha D^\alpha}{\rho_0 J}\right)^{-1/\alpha}$	$-2\alpha \ln 2$

Mitchell, Vojta, Bulla, Fritz, PRB 88, 195119 (2013)

A van Hove peak boosts T_K

	Density of states $\rho(\omega)$	T_K	$S_{\rm imp}(T \to 0)$
Metal	ρ_0 : constant	$D\, e^{-1/\rho_0 J}$	0
Dirac cone	$\rho_0 \lvert\omega\rvert$: linear pseudogap	$T_K = 0$	$\ln 2$
Standard van Hove	$\rho_0 \left[1 + a \ln(D/\lvert\omega\rvert)\right]$: log divergence	$D\, e^{\frac{1}{a} - \frac{1}{a}\sqrt{1+\frac{2a}{\rho_0 J}}}$	0
Higher-order van Hove	$\rho_0 \lvert\omega\rvert^{-\alpha}$: power-law divergence	$D \left(1 + \frac{\alpha D^\alpha}{\rho_0 J}\right)^{-1/\alpha}$	$-2\alpha \ln 2$

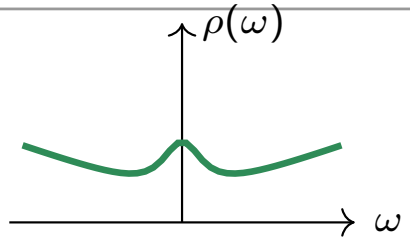
Mitchell, Vojta, Bulla, Fritz, PRB 88, 195119 (2013)

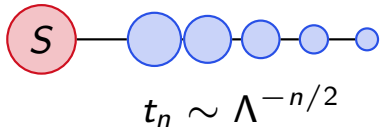
Logarithmic vHs: T_K enhanced over the metal

Higher-order vHs: T_K grows as a power law in J

In TBG the cone and the vHs sit within a few meV of each other: which one wins?

NRG settles it non-perturbatively: read the answer off the impurity entropy

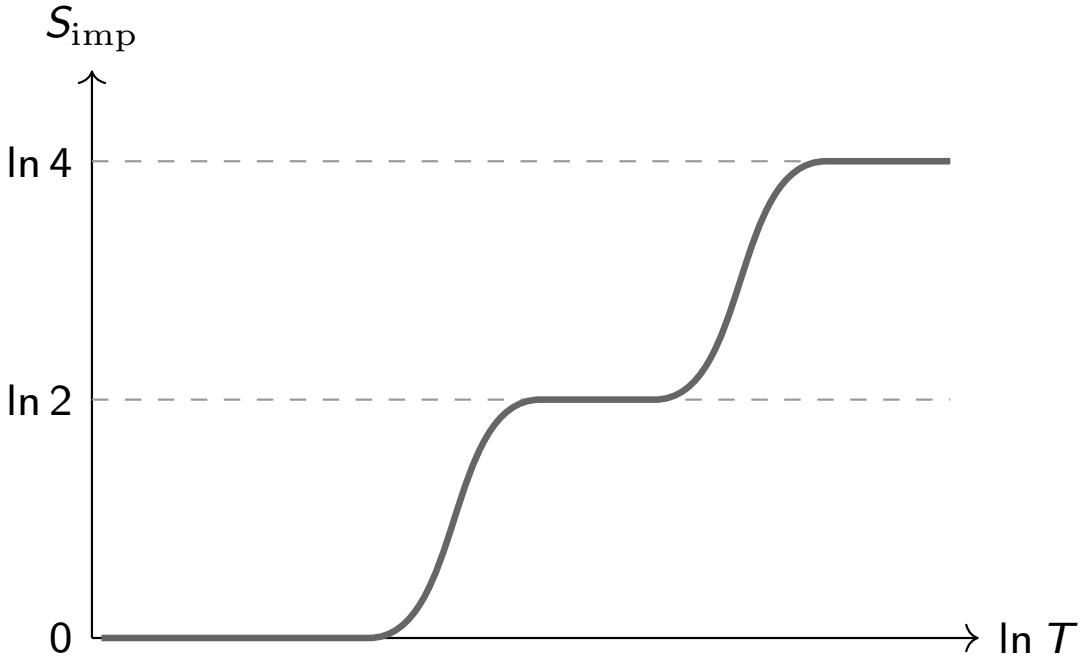




3. $S_{\rm imp}(T)$

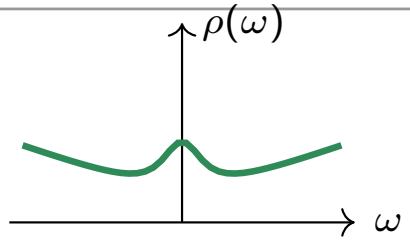
1. host DoS $\rho(\omega)$

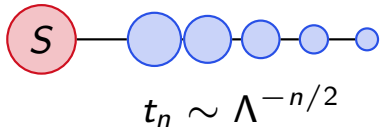
2. Wilson chain, solved iteratively



A plateau at ln **N**: **N states** available to the impurity

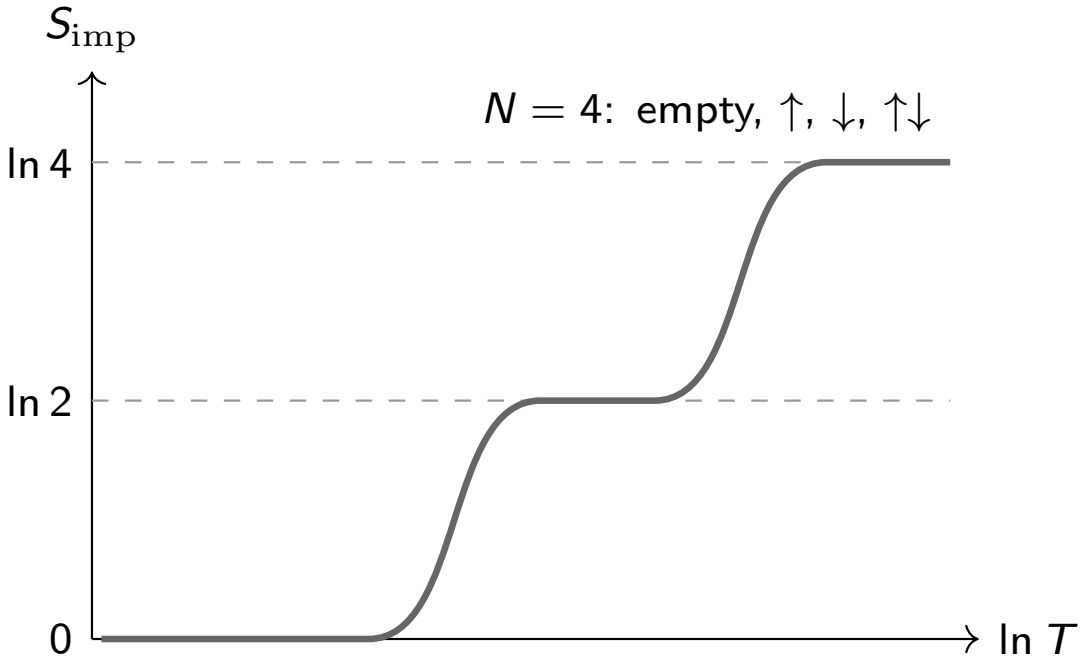
NRG settles it non-perturbatively: read the answer off the impurity entropy





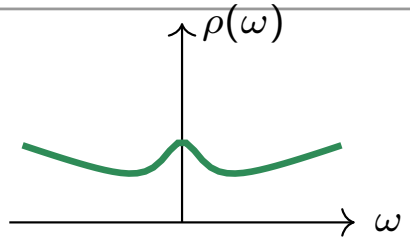
3. $S_{\rm imp}(T)$

1. host DoS $\rho(\omega)$ **2.** Wilson chain, solved iteratively

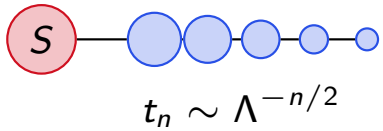


A plateau at ln **N**: **N states** available to the impurity

NRG settles it non-perturbatively: read the answer off the impurity entropy

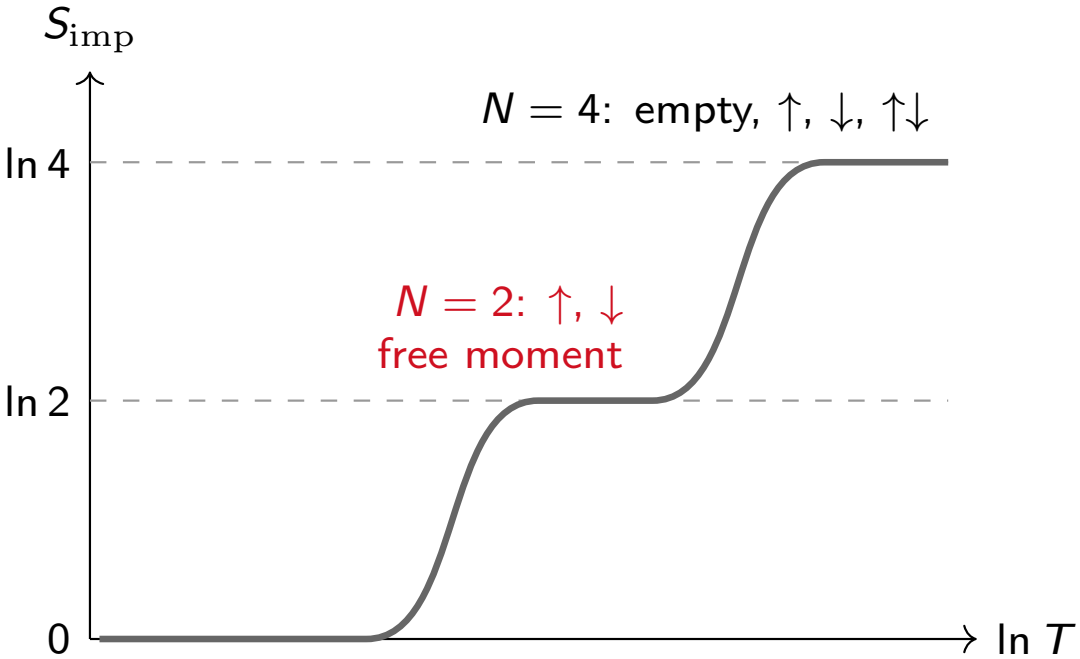


1. host DoS $\rho(\omega)$



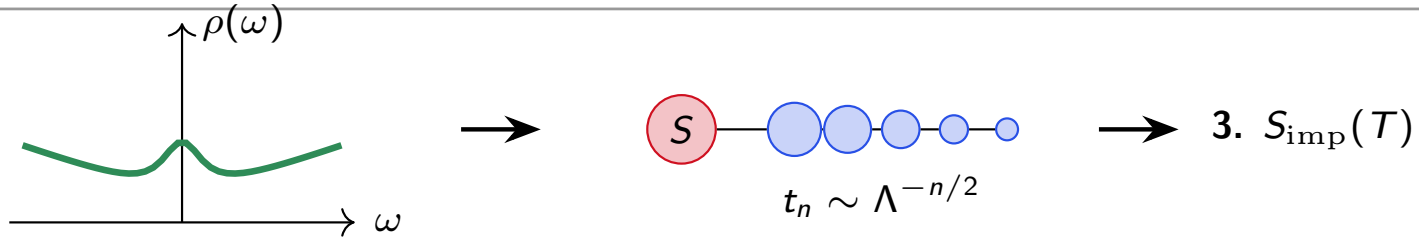
2. Wilson chain, solved iteratively

3. $S_{\mathrm{imp}}(T)$

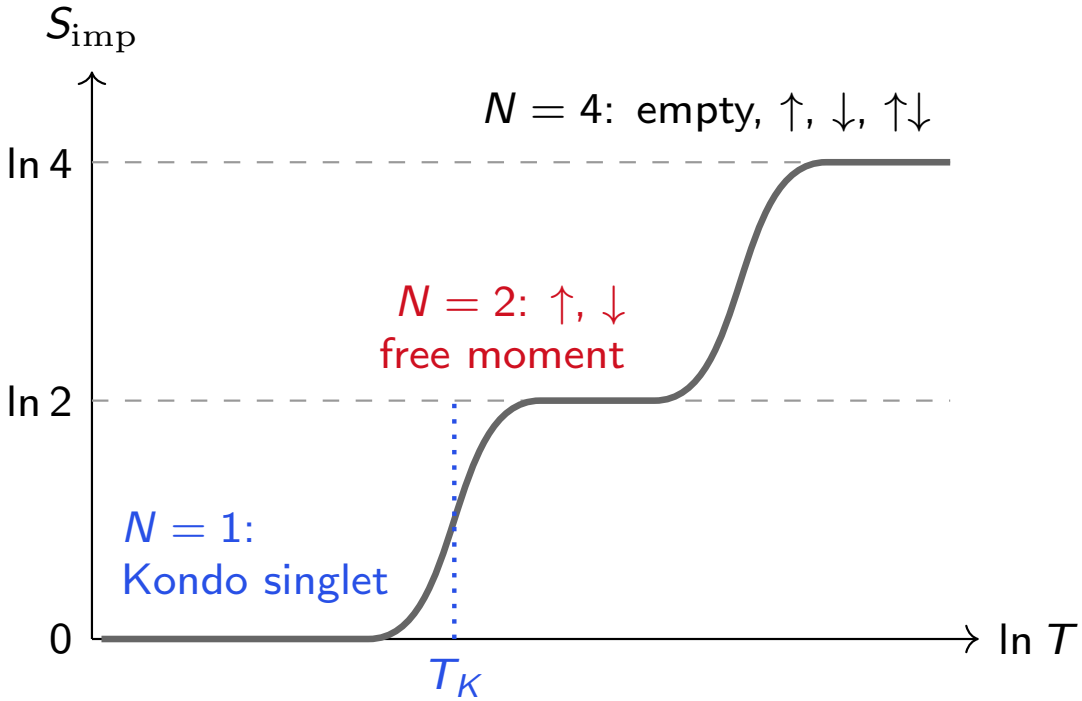


A plateau at ln **N**: **N states** available to the impurity

NRG settles it non-perturbatively: read the answer off the impurity entropy



1. host DoS $\rho(\omega)$ **2.** Wilson chain, solved iteratively



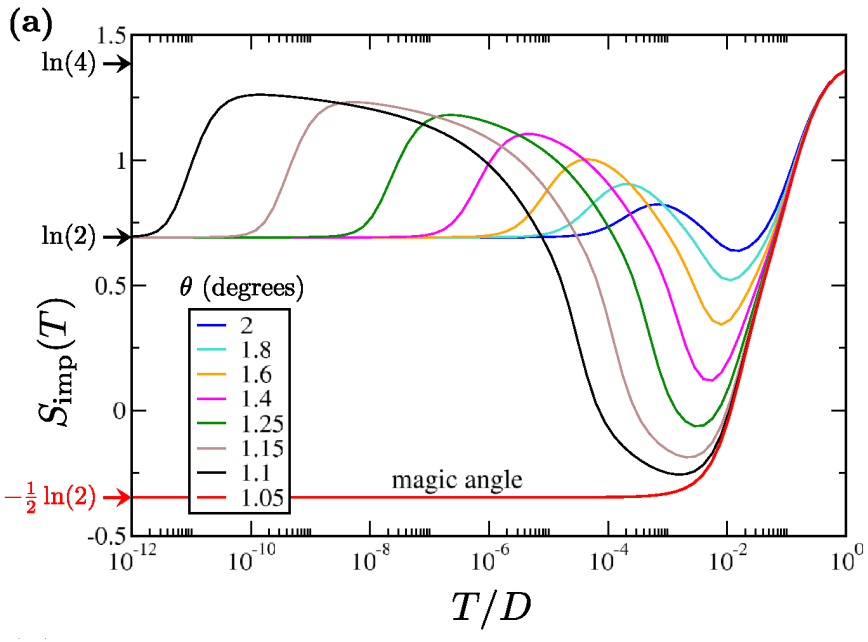
A plateau at ln **N**: **N states** available to the impurity

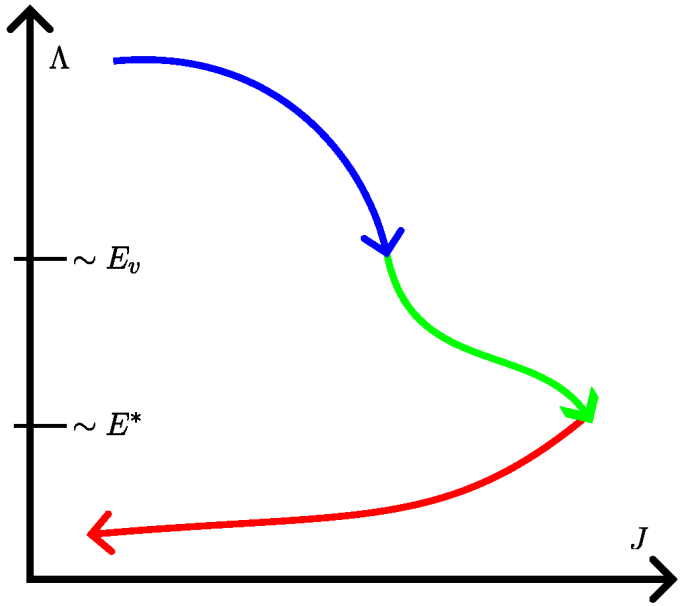
A step between plateaus: a screening scale, here T_K

$S_{\rm imp} = S_{\rm total} - S_{\rm host}$, so it can be negative

This curve is the one diagnostic used for the rest of the talk

Kondo revival at the magic angle



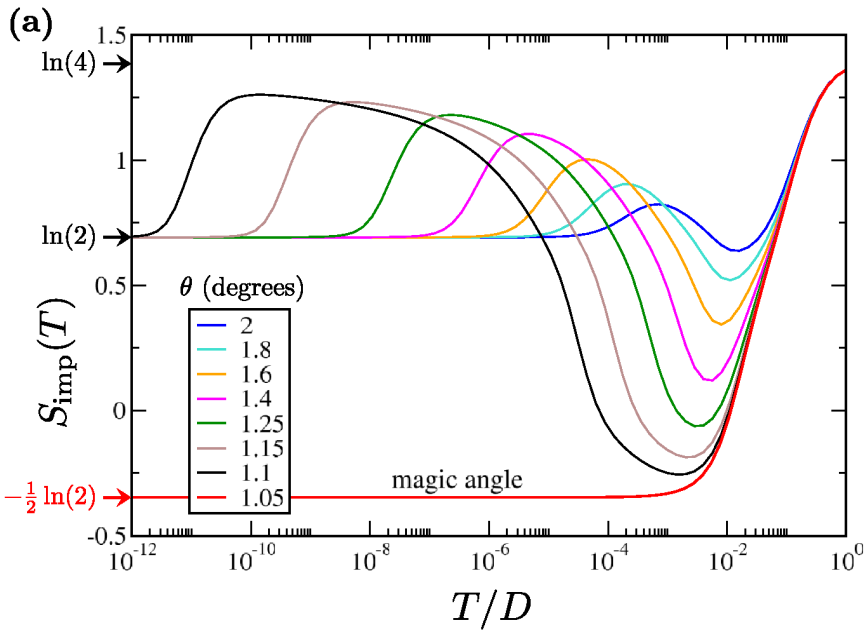


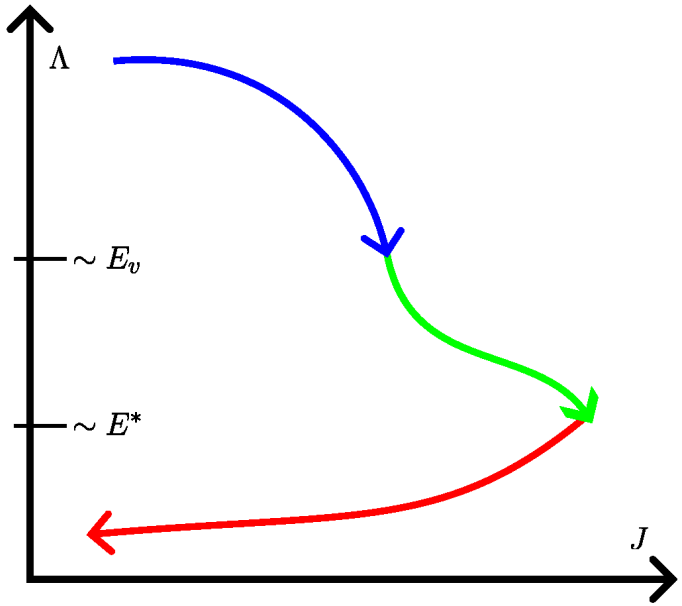
Shankar, Oriekhov, Mitchell, Fritz, PRB 107, 245102 (2023)

See also: Chang, Yi, Wu, Kugler, Andrei, Vanderbilt, Kotliar, Pixley, PRL 133, 126503 (2024)

Screening begins at E_v; below $E^* \sim |E_v|^3$ the cone wins: moment unscreened

Kondo revival at the magic angle





Shankar, Oriekhov, Mitchell, Fritz, PRB 107, 245102 (2023)

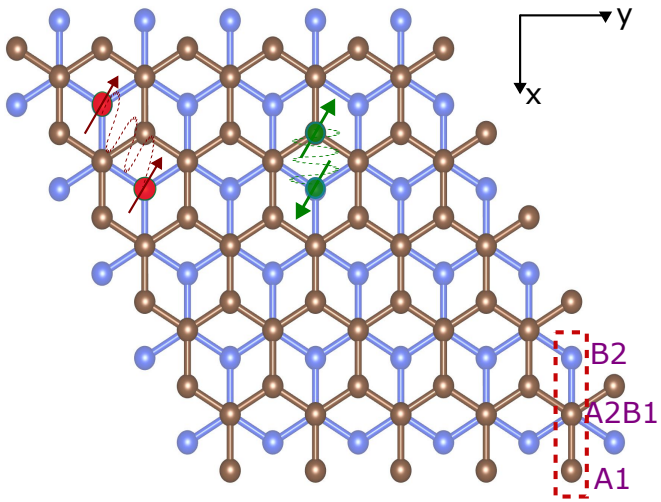
See also: Chang, Yi, Wu, Kugler, Andrei, Vanderbilt, Kotliar, Pixley, PRL 133, 126503 (2024)

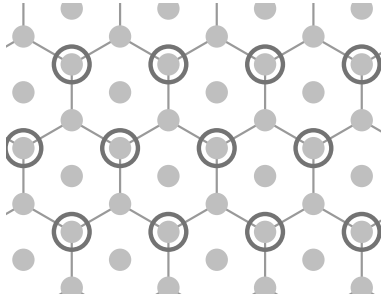
Screening begins at E_v; below $E^* \sim |E_v|^3$ the cone wins: moment unscreened

Magic angle: E_v, $E^* \to 0$, the cone never wins: moment screened

One impurity is only the dilute limit. What happens when two sit close together?

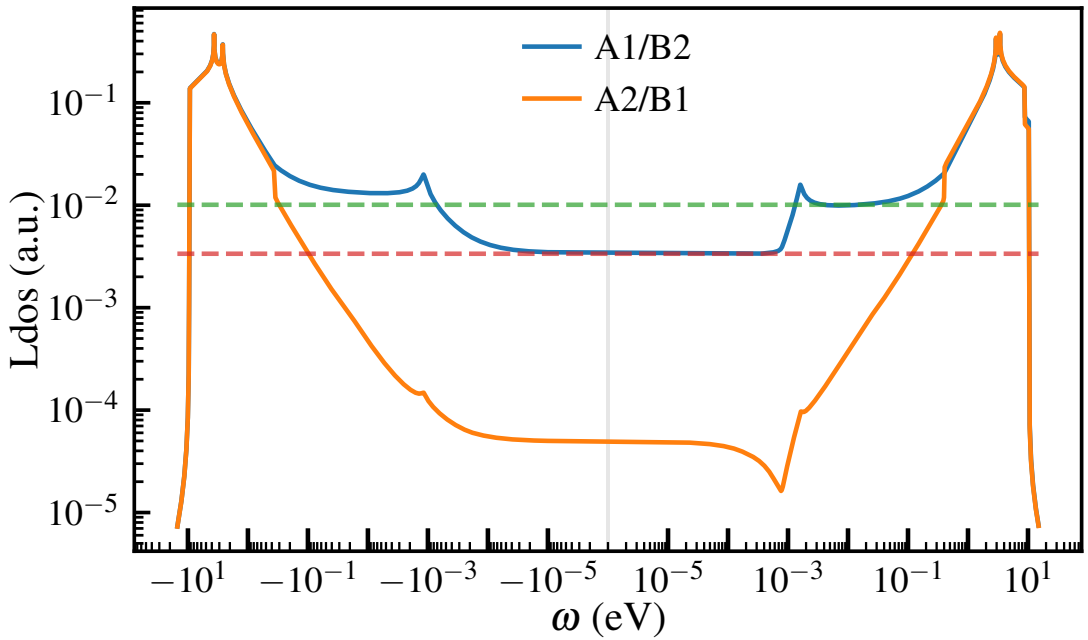
Bernal bilayer: stacked and unstacked sites see very different DoS





unstacked: A1, B2

stacked: A2/B1

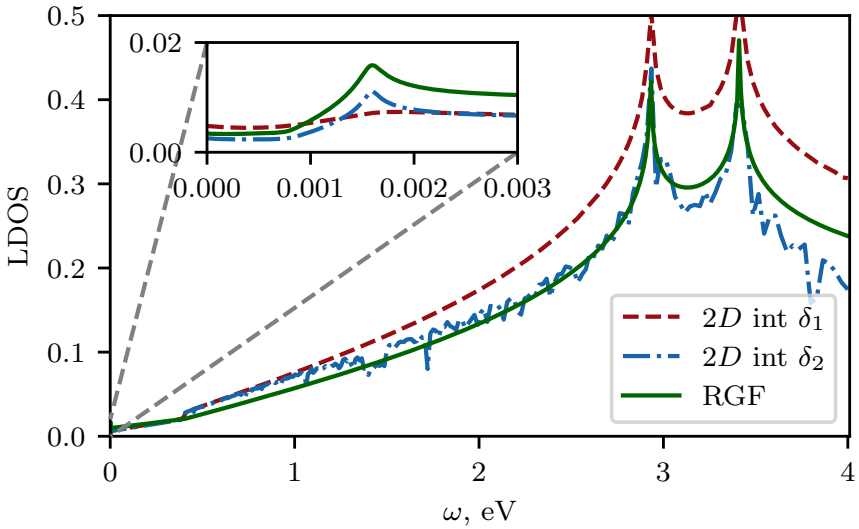


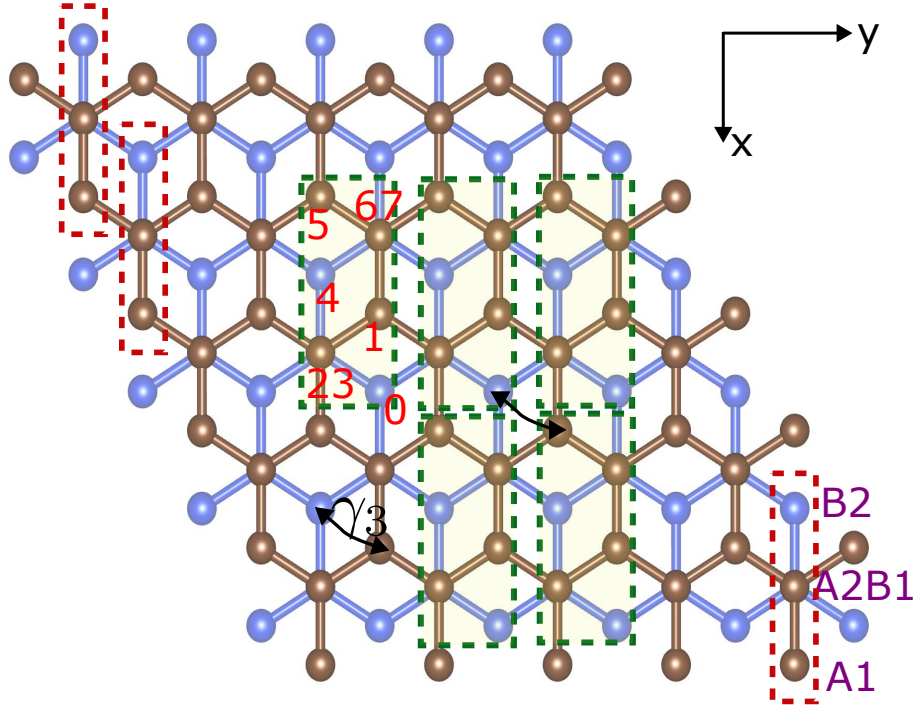
A1/B2 (unstacked): large LDoS, trigonal-warping vHs at $\sim$1.6 meV

A2/B1 (stacked): suppressed LDoS at low energy

Same lattice, two kinds of site: what does an impurity feel?

RGF resolves the 1.6 meV vHs that 2D integration misses





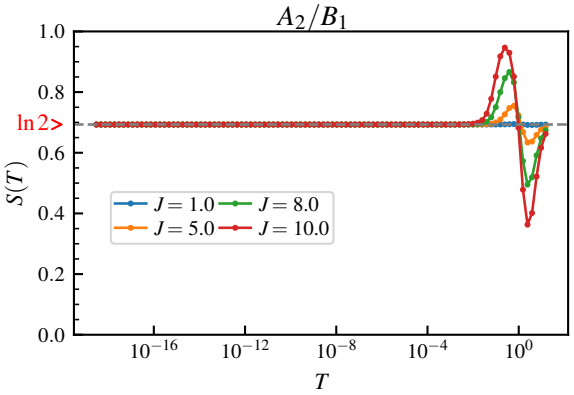
vHs lives within 10^{-4} of the BZ scale: 2D integration needs a huge grid

Fold into a square supercell, Fourier transform along one axis, recurse

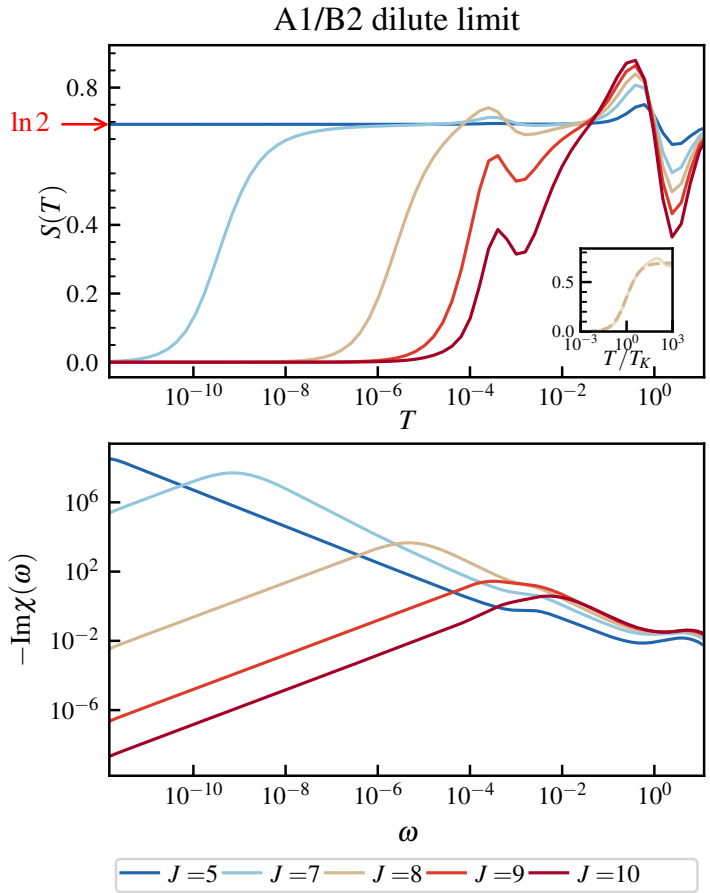
RGF gives the full energy-resolved local and non-local DoS cheaply

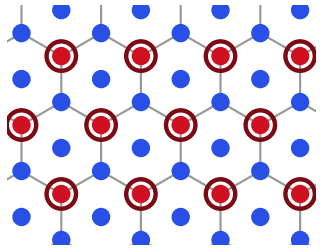
Stacked sites never screen; unstacked sites do

A2/B1



A1/B2





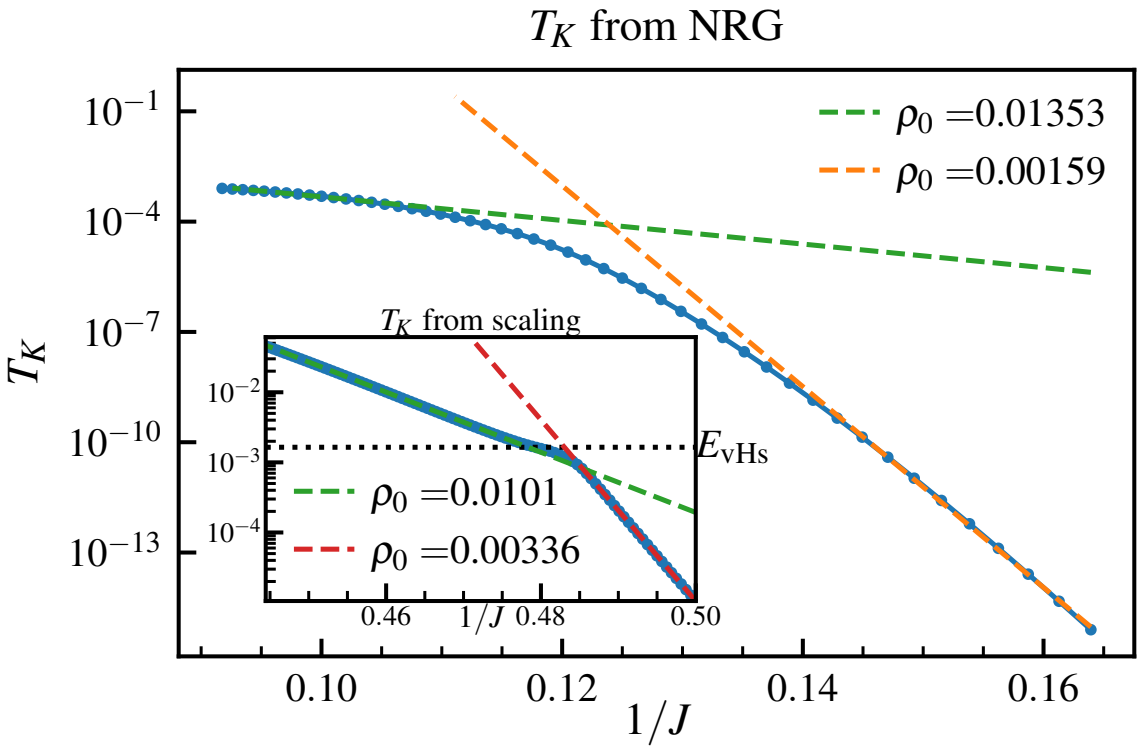
- free moment (A2/B1)
- Kondo (A1/B2)

A2/B1: $S \to \ln 2$

A1/B2: $S \to 0$ at large J

Free moment vs Kondo singlet is set by the impurity position alone

T_K crosses between two DoS plateaus as the coupling grows



Weak J: low-energy DoS plateau

Strong J: flow reaches the high-energy plateau

Smooth crossover, no transition

$$T_K \sim D \exp\left(-1/\rho_{\mathrm{eff}} J\right)$$

Poor man's scaling captures it qualitatively; NRG confirms it quantitatively

Two impurities: the RKKY sign follows the sublattice

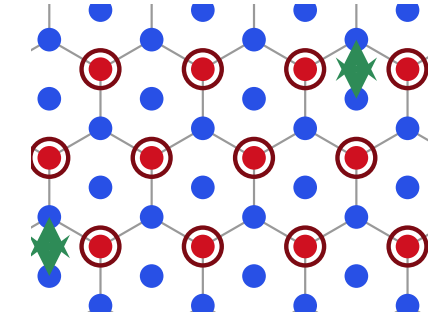
$$H = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} c_{\mathbf{k}}^{\dagger} c_{\mathbf{k}} + J \sum_{i=1,2} \mathbf{S}_i \cdot \boldsymbol{\sigma}(\mathbf{r}_i)$$

Same sublattice: **FM**

Opposite sublattice: **AFM**

$J_{\mathrm{RKKY}} \propto J^2$, $T_K \sim e^{-1/J}$

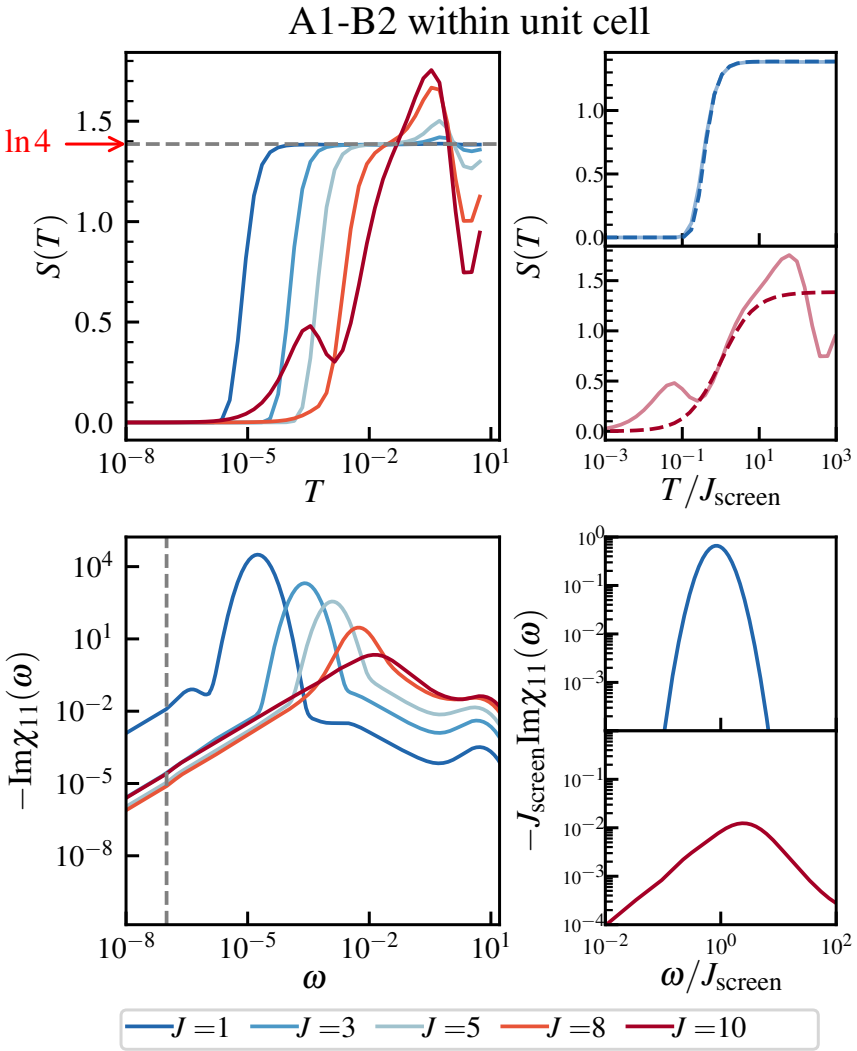
pair	competing regimes
A2–A2	FM / LM
A2–B1	AFM / LM
A1–B2	AFM / Kondo
B2–B2	FM / Kondo



- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)

Only A1–B2 and B2–B2 have a nontrivial RKKY–Kondo competition

A1–B2: an AFM singlet at weak J crosses over to Kondo screening



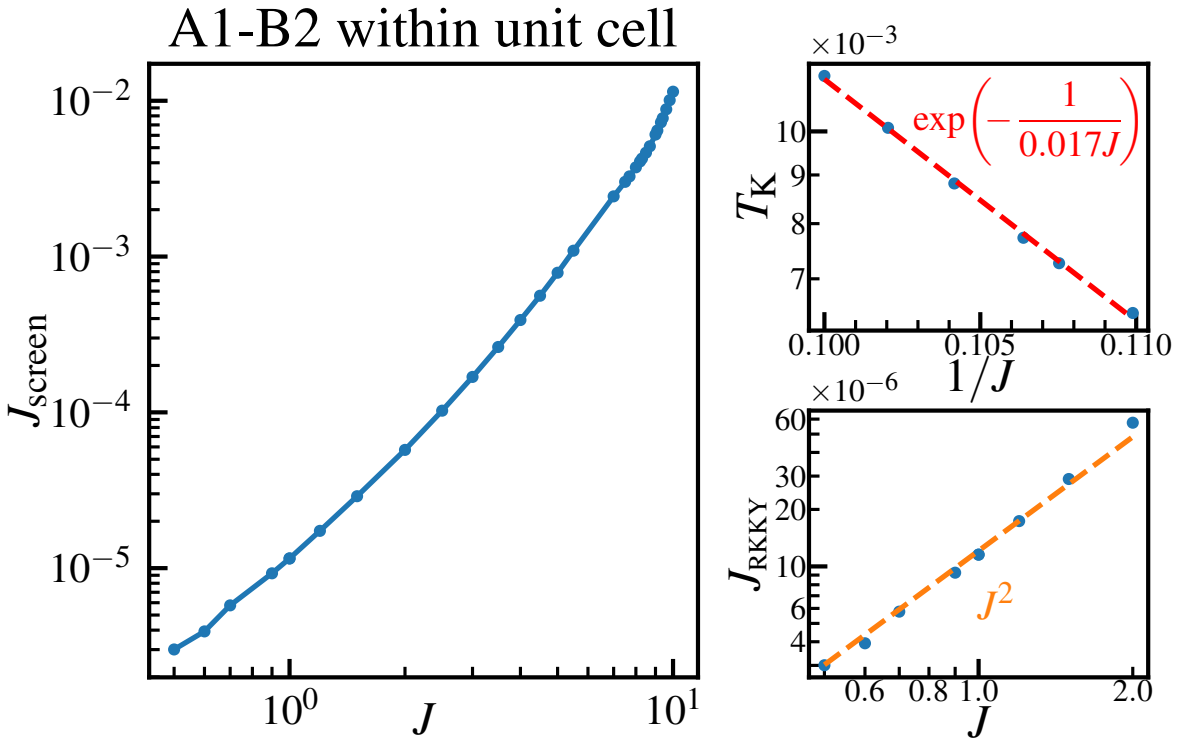
Weak J: entropy $\ln 4 \to 0$ via $\ln 2$, effective two-spin model $J_{\mathrm{RKKY}}\mathbf{S}_1\cdot\mathbf{S}_2$

Strong J: two independent Kondo screenings

Both limits follow universal curves

Antiferromagnetic locking and Kondo screening compete at one scale

The screening scale goes from J^2 (RKKY) to $e^{-1/J}$ (Kondo)



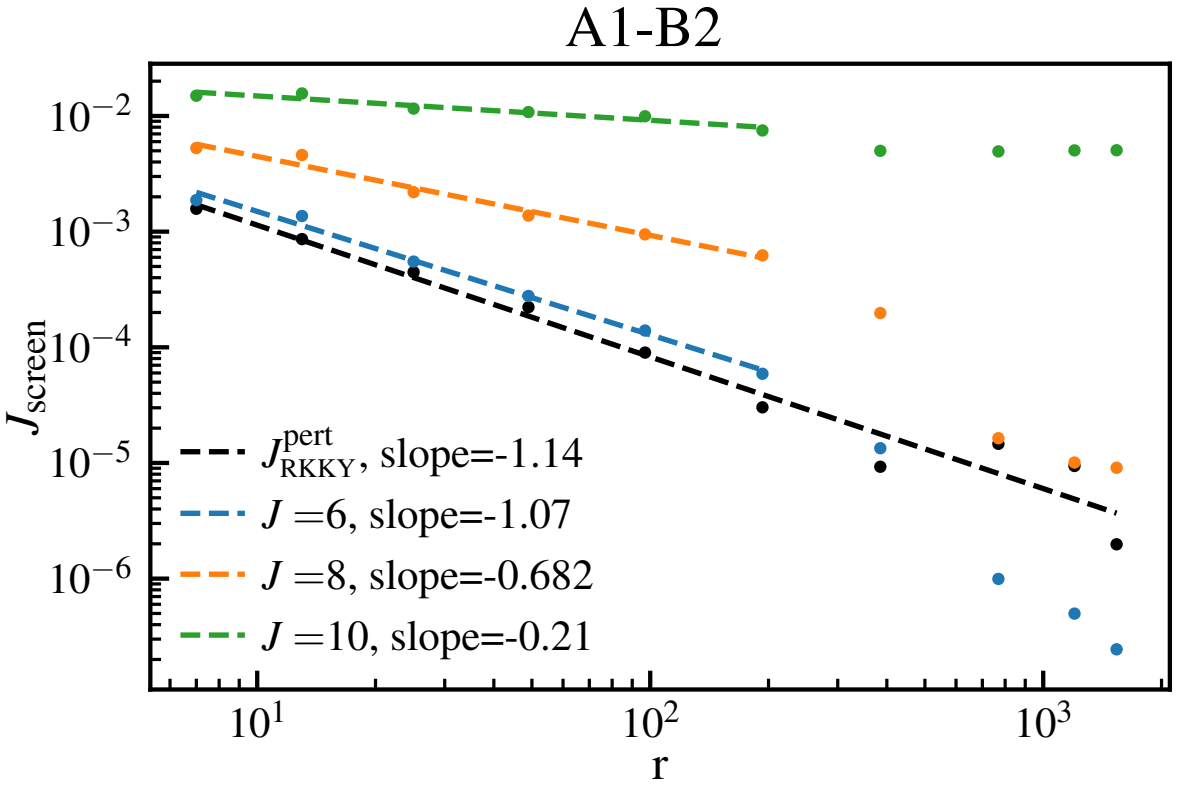
Weak J: $J_{\mathrm{screen}} \propto J^2$

Strong J: $\propto e^{-1/J}$

Insets confirm both laws

Same configuration, two regimes: the competition is non-perturbative

The distance dependence flattens from -1.14 to -0.21 with coupling



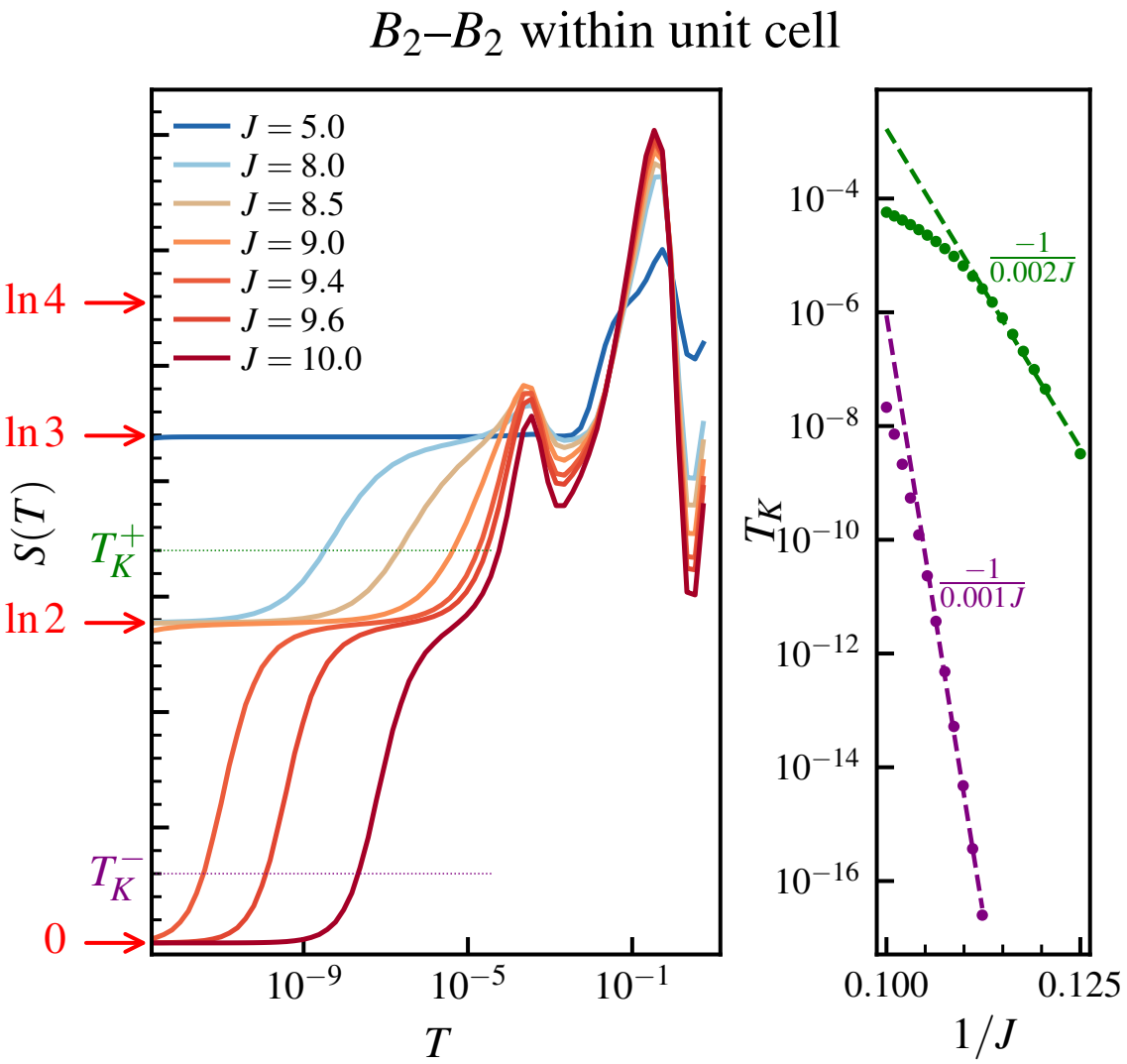
Perturbative RKKY: slope -1.14

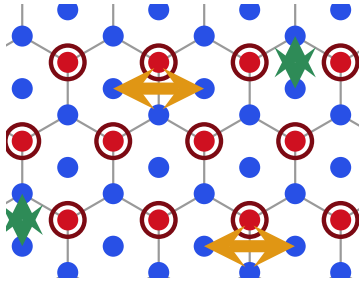
$J = 10$: slope -0.21

Strong coupling approaches the dilute Kondo limit

At strong coupling the impurities barely feel each other

B2–B2: an FM lock leads to two-stage Kondo screening





- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)
- FM two-stage (B2–B2)

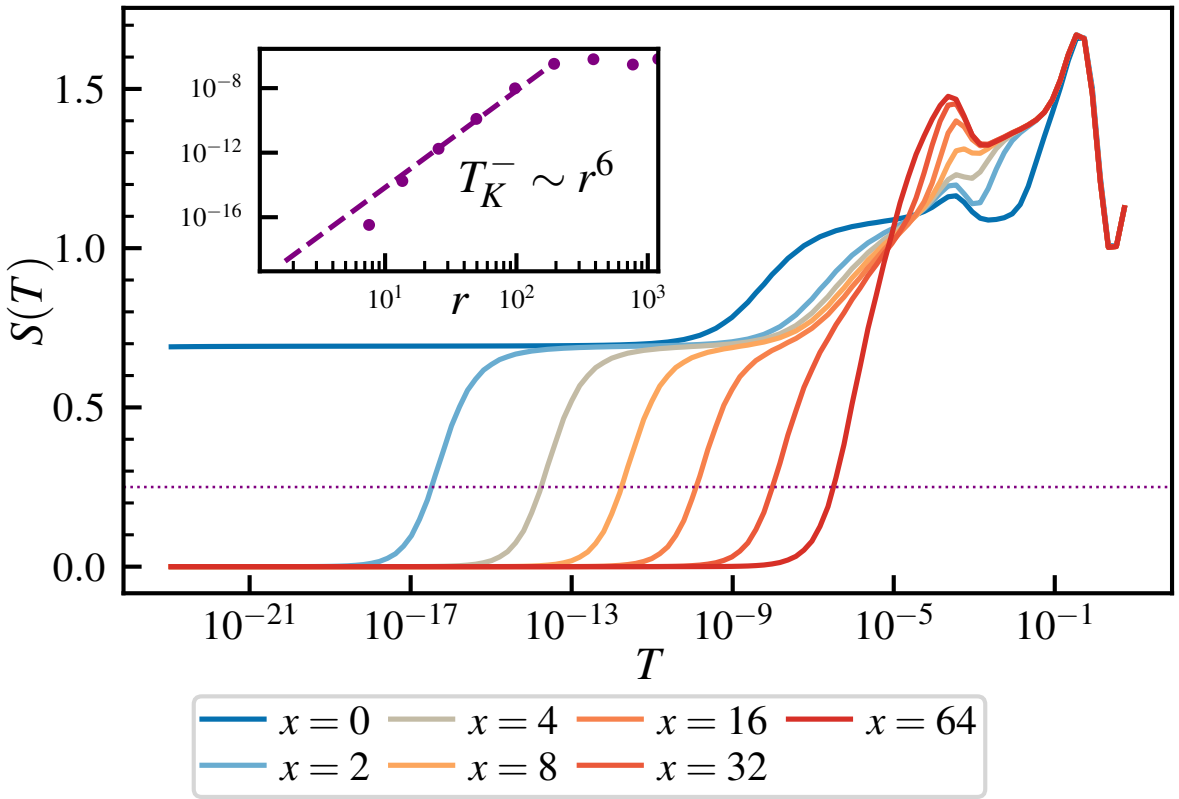
$\ln 4 \to \ln 3$ (FM spin-1)

$\to \ln 2$ (underscreened, T_K^+)

$\to 0$ (singlet, T_K^-)

Entropy plateaus: $\ln 4 \to \ln 3 \to \ln 2 \to 0$

With distance the two Kondo scales merge into the dilute limit



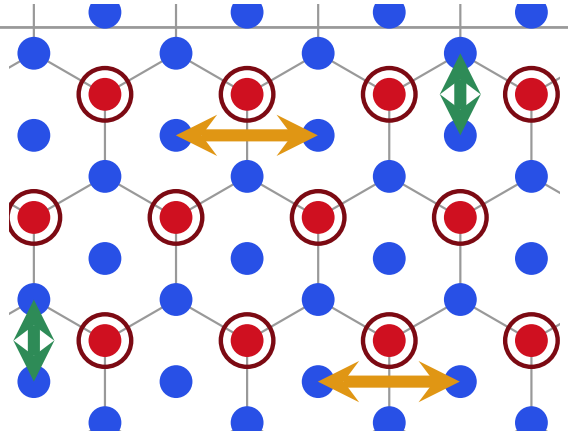
$T_K^{\pm}$ approach each other as r grows

Two-stage structure needs a sizeable FM RKKY

Dilute limit recovered smoothly

Two-stage screening is strongest for close impurities

One host, every screening regime: the impurity position selects it



- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)
- FM two-stage (B2–B2)

Stacked: free moment, never screened

Unstacked: Kondo, AFM singlet, FM two-stage

Position and pairing pick the regime

Outlook: STM on BLG, artificial impurity arrays, RKKY in TBG

Site selectivity: every Kondo regime in one sample

Backup: universal two-spin entropy curves

Antiferromagnetic, $J > 0$: $S(J{=}T) = \ln 3 + \ldots$ (universal function of $u = J/4T$)

Ferromagnetic, $J < 0$: spin-1 plateau, $S \to \ln 3$

Kondo: $S(T) \simeq \ln 2/(1 + T_K/T)$, with $S(T_K) = \frac{1}{2} \ln 2$

Backup: perturbative RKKY, sign from the NLDOS

$$\rho(\mathbf{r},\omega) = \rho_s(\mathbf{r},\omega) + \rho_a(\mathbf{r},\omega), \quad \rho_{s/a}(-\omega) = \pm\rho_{s/a}(\omega)$$

$J_{\mathrm{RKKY}} \propto J^2$ from the one-loop spin susceptibility

Positive denominator: sign set by the symmetric vs antisymmetric NLDOS

Same sublattice: symmetric dominates, FM; opposite: AFM

Backup: toy model, RKKY develops a logarithmic decay

